Qiao Sun

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RESEARCH INTERESTS

I seek to understand how learning and computation give rise to intelligence. My primary interests lie in foundational AI research, particularly training algorithms.

RESEARCH EXPERIENCE

He Vision Group

Sep 2024 – Jul 19, 2026

Undergraduate Researcher · Professor: Kaiming He

- Conducted research on generative modeling, fast image generation, visual understanding, and AI agents.
- Served as one of the group leads, helping coordinate compute resources, access, and shared research infrastructure.

Google

Jul 20, 2026 – Present

Student Researcher · Host: Deqing Sun

- Ongoing research on optimization for recurrent methods and looped models.

SELECTED PAPERS

Is Noise Conditioning Necessary for Denoising Generative Models?

2025

ICML 2025 · Poster · First author

- Revisited the necessity of noise conditioning in diffusion and flow-matching models by reimplementing eight denoising generative models across multiple datasets.
- Proposed uEDM, a noise-unconditional diffusion model with competitive performance, supported by theory that matches empirical behavior.

Bidirectional Normalizing Flow: From Data to Noise and Back

2025

CVPR · Spotlight · Project lead and first author

- Revisited normalizing flows with a learned reverse map, eliminating explicit inverse-flow computation and slow autoregressive inference.
- Guided reverse learning through hidden-state alignment, enabling single-evaluation NF-based generation with state-of-the-art results among NF methods.

One-step Latent-free Image Generation with Pixel Mean Flows

2026

ICML · Accepted · First author

- Built a strong baseline for one-step, pixel-space, latent-free generation using MeanFlow with x-prediction.
- Reported 2.22 FID on ImageNet 256 and 2.48 FID on ImageNet 512; open-source implementation at Lyy-iiis/pMF.

Qiao Sun

EDUCATION

Massachusetts Institute of Technology Sep 2024 – Present
Undergraduate in Physics and Mathematics
Expected graduation: 2028 · GPA: 4.9/5.0

Tsinghua University Sep 2023 – Jul 2024
Institute for Interdisciplinary Information Sciences
Pre-college student · GPA: 4.00/4.00

PROFESSIONAL EXPERIENCE

Ubiquant Investment (Jiukun Quant) Jun 20 – Aug 31, 2025
Quant Strategy Analyst Intern · Quantitative research and strategy analysis

- Researched quantitative strategy ideas and backtesting workflows with Python-based data analysis.
- Evaluated factor robustness, risk and turnover characteristics, and strategy diagnostics through statistical tests and visualization.
- Built reproducible research utilities for experiment tracking, factor validation, and portfolio-performance analysis.

COURSE PROJECTS

Fast Humanoid Loco-Manipulation via Flow Matching 2025
MIT 6.4210 · Robotics manipulation · Project video

- Reimplemented a BeyondMimic-style humanoid loco-manipulation pipeline with simulated data preparation, diffusion training, and post-hoc control guidance.
- Replaced DDPM sampling with flow matching and demonstrated lower-latency control with 5 sampling steps.

HONORS & AWARDS

Top 17 · Putnam Mathematical Competition 2025

2nd place · Putnam Mathematical Competition 2024

Gold Medal, 11th place · International Mathematical Olympiad 2023

Gold Medal, 1st place with perfect score 2022
Chinese Mathematical Olympiad

Excellent Award · Alibaba Global Mathematics Competition 2022–2024
Top 70 among 50,000+ participants

SKILLS

Programming & AI
Proficient in general-purpose programming and the design, implementation, and evaluation of AI algorithms.

Languages Mandarin (native), English (fluent)